CREDIT CARD FRAUD DETECTION

Submitted By:

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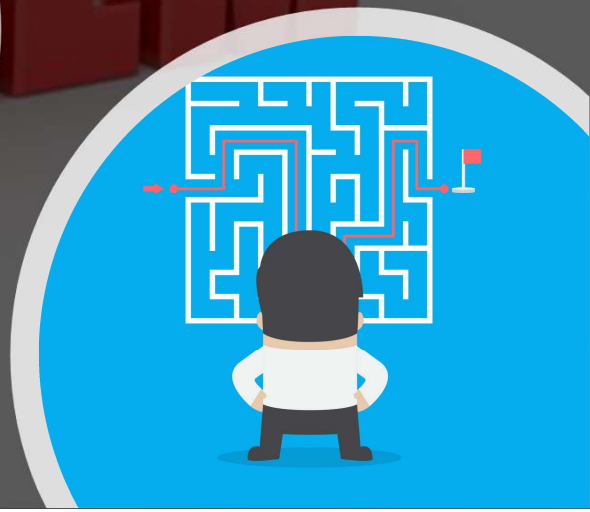
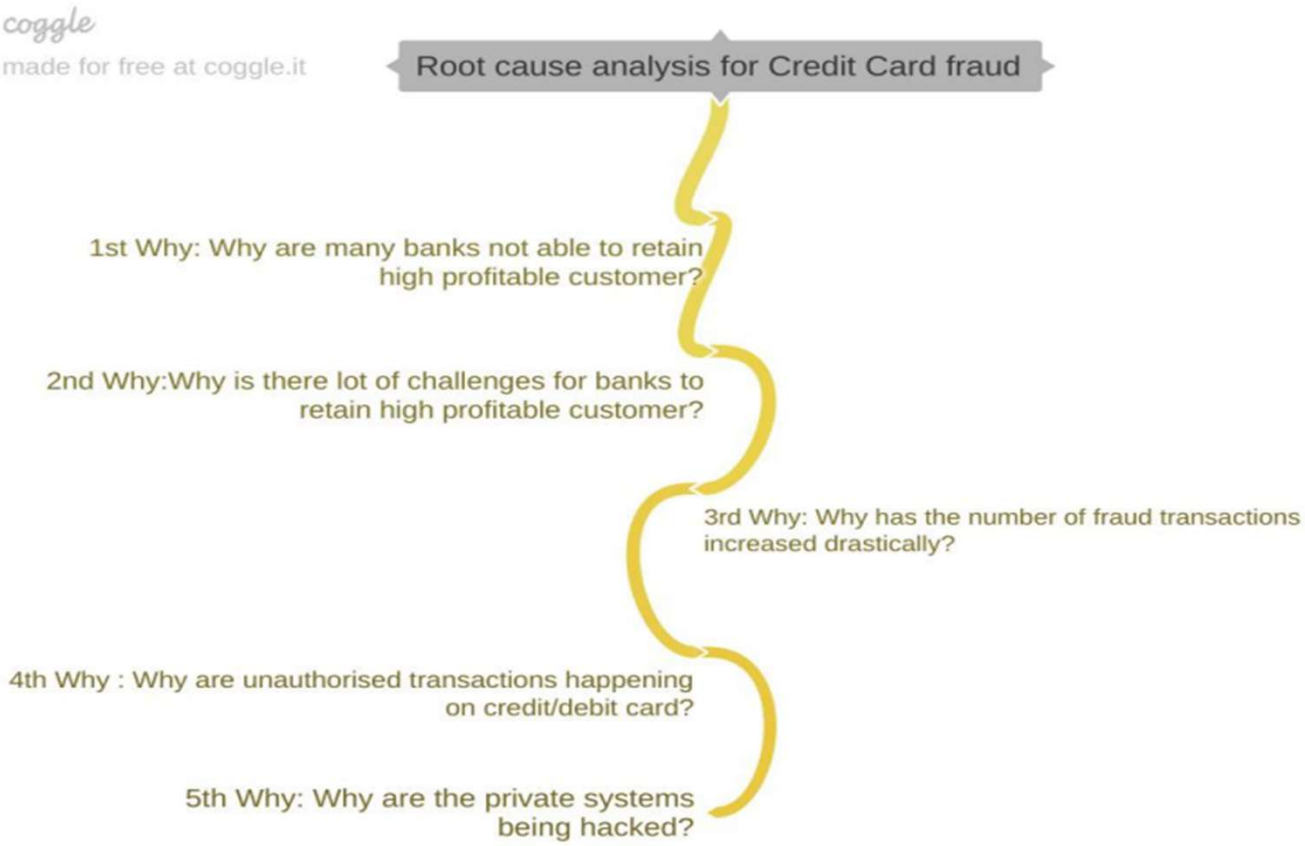
BACKGRAOUND OF PROBLEM

- Huge revenue loss due to unauthorised transactions during non-peak and odd hours of the day.
- Source: Stolen/lost cards, ATM skimming at various POS terminals.
- Stopping the fraudsters/fraudulent activity is highly impossible as they actively find alternate ways.
- Extra level authentication is complex & tedious job.
- Goal is to detect and prevent the fraudulent transactions before the financial crisis.



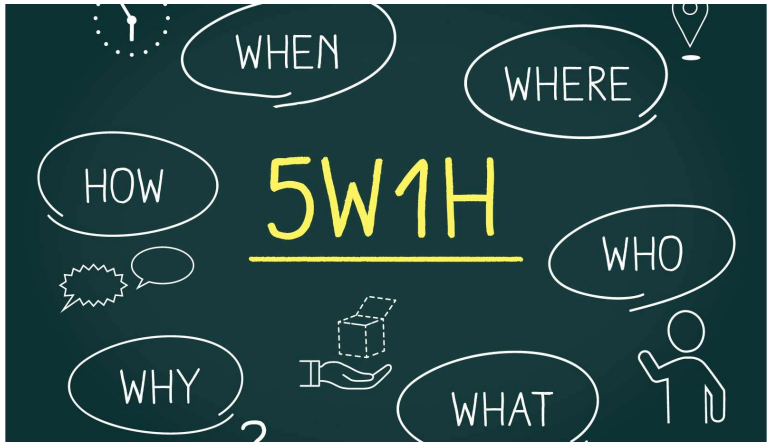
ROUTE CAUSE ANALYSIS

- 5 WHYs
- Sensitive credit card details of customers is being accessed.
- Private systems are hacked, stolen/lost cards are misused, ATM skimming at POS terminals.



UNDERSTANDING THE PROBLEM & IMPACT

1. Who is involved in the process?
    - Fraudsters, skimmers who get unauthorized access to a customer's credit card info.
2. What do they do with it?
    - Commit fraudulent transactions: alter genuine cards, create counterfeit cards, steal credit cards, fraudulent telemarketing.
3. Where do the transactions happen?
    - Most of them happen online and sometimes offline.
    - Online frauds are hard to detect due to difficulty in amassing evidence, time & resource constraints.
4. When does it happen?
    - Most happen at non-peak and odd hours of the day when user in-active or asleep.
5. How is business affected by this?
    - As bank need to repay customer the total transaction amount in case of fraudulent transaction
    - Banks suffer from substantial financial loss, lose trust & customer's credibility.

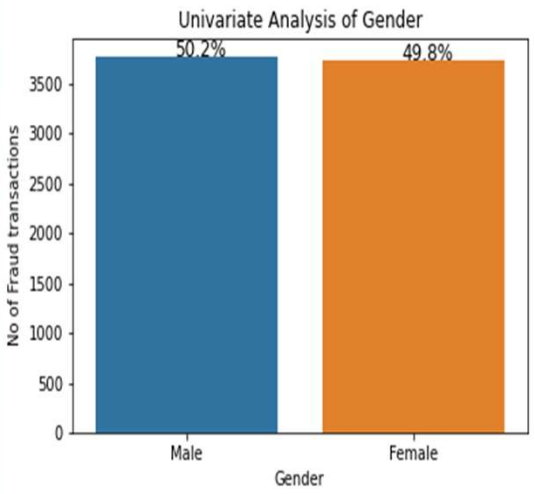


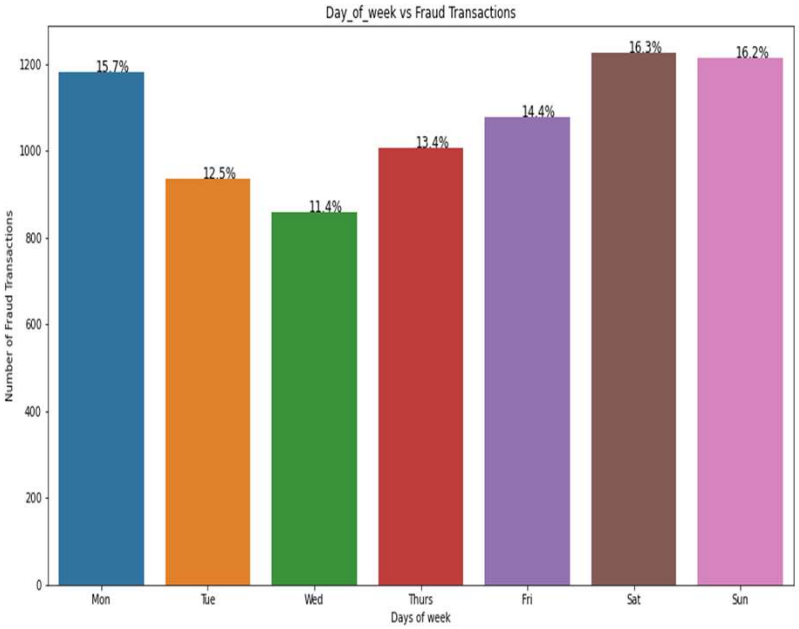
PROCESS FLOW

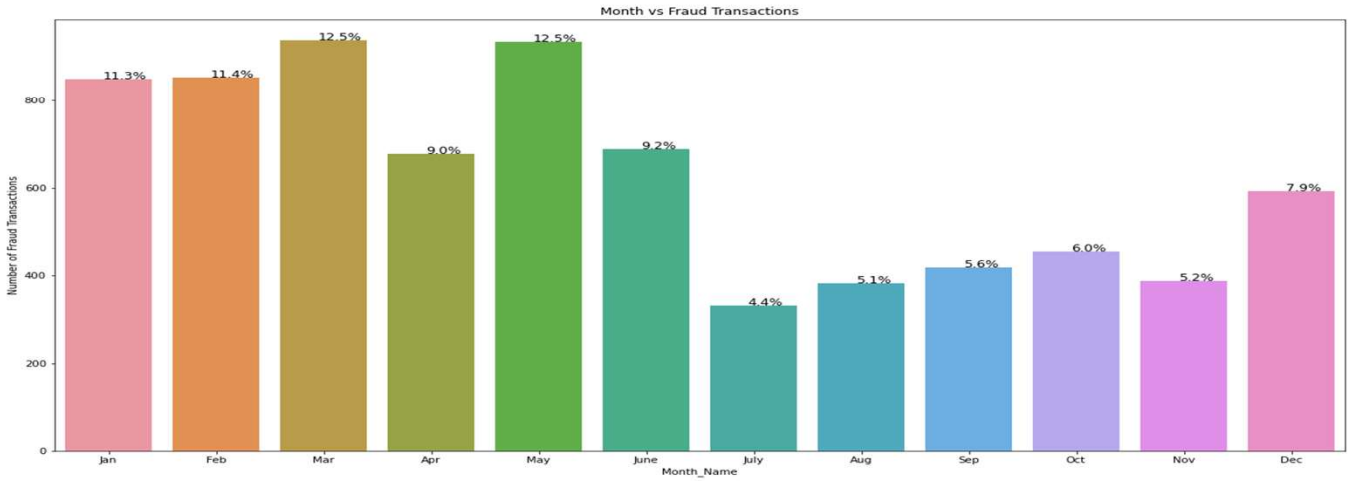
Data Import & EDA on Train Data → Data Cleaning → Selecting Feature with significant impact → Train, Test Split → Build models with various techniques → Validate model with Recall & AUC → Compare model performance & chose the best → Model Fitting on Test/unseen Data → Validate model with Recall & AUC → Cost Benefit analysis

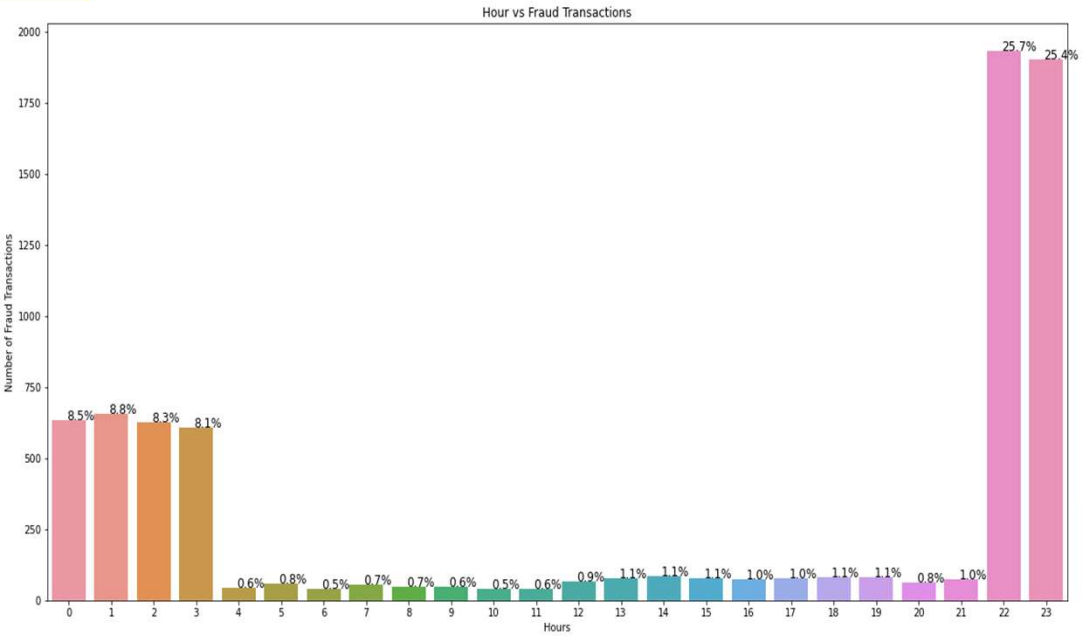
CHARACTERISTICS OF FRAUDULENT TRANSACTIONS

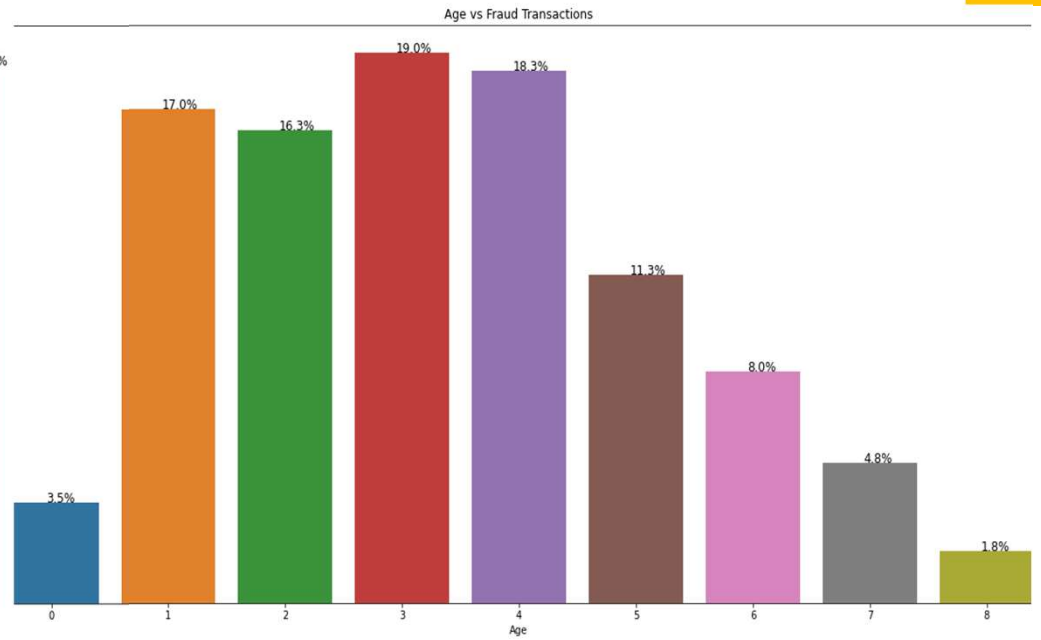
- Gender v/s fraudulent transactions
- Day of the week v/s fraudulent transactions
- Month v/s fraudulent transactions











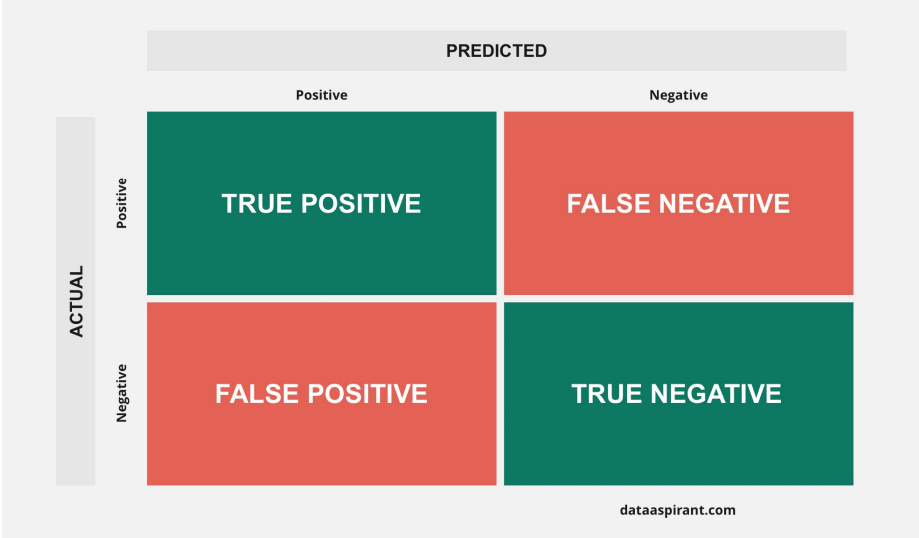
CHARACTERISTICS OF FRAUDULENT TRANSACTIONS

- Hours of day v/s transactions
- Age v/s fraudulent transactions

MODEL PERFORMANCE ON TRAINING DATA

Evaluation metrics: Recall, AUC

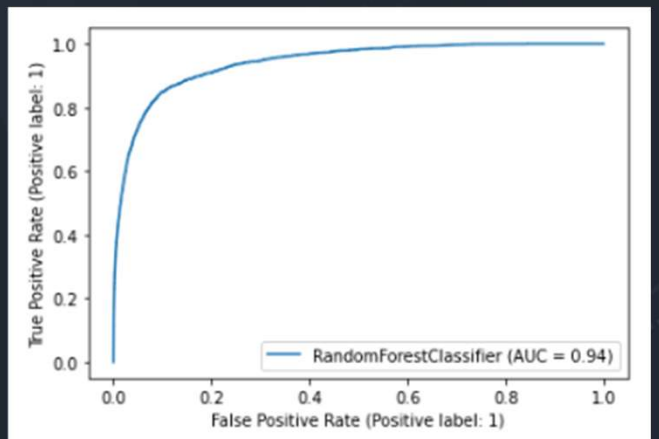
1. Why Recall?
   - Most likely to predict actual Positive cases(Fraud transactions).
   - Very helpful to reduce financial loss.
   - Recall = TP / (TP + FN)
   - High Recall refers to low false negative score.
2. Why AUC?
   - Best to measure overall performance of a binary classification problem.
   - Best to select optimal threshold value of a model.
   - Plots TPR v/s FPR.



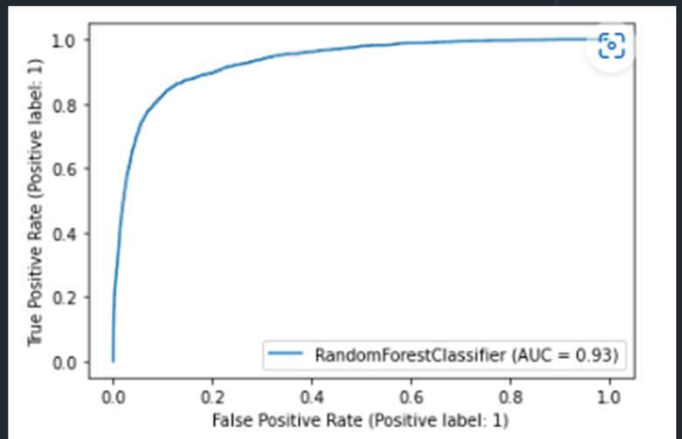
	Model Name	Train_Recall	Test_Recall	AUC__Scor
3	Decision Trees - Imbalanced	0.24	0.23	0.9
6	Random Forest - Imbalanced	0.00	0.00	0.9
7	Random Forest - SMOTE	0.87	0.72	0.8
8	Random Forest- ADASYN	0.86	0.70	0.8
5	Decision Trees- ADASYN	0.92	0.82	0.5
4	Decision Trees - SMOTE	0.92	0.88	0.5
1	Logistic Regression - SMOTE	0.59	1.00	0.5
0	Logistic Regression - Imbalanced	0.00	0.99	0.4
2	Logistic Regression - ADASYN	0.58	1.00	0.4

MODEL PERFORMANCE ON TEST/UNSEEN DATA

- Trade off b/w Random Forest SMOTE & Random Forest ADASYN.
  - Still choose Random Forest ADASYN why?
  - Impact from 1% increase in Recall score.
  - 1% of overall transaction = 18,524 (Train+Test)
- Assumption: Financial impact of 1% fraud in 1% of overall transaction
  - 1% of Fraudulent transactions = 0.01% overall transaction = 185 transactions.
  - Avg amt for 185 transaction = (avg amount/transaction) * $185 = $98,050



Random forest SMOTE - Metrics : Recall :0.86 AUC :0.94



Random forest ADASYN - Metrics : Recall :0.87 AUC :0.93

COST BENEFIT ANALYSIS

Cost Benefit Analysis		
S. No	Questions	Answer
a	Average number of transactions per month	77183.00
b	Average number of fraudulent transaction per month	402.00
c	Average amount per fraud transaction	530.00

Cost Benefit Analysis		
S. No	Questions	Answer
1	Cost incurred per month before the model was deployed = (avg no of fraud transaction) * (avg amt / fraud transaction)	$2,13,060.00
2	Average number of transactions per month detected as fraudulent by the model (TF)	28
3	Cost of providing customer executive support per fraudulent transaction detected by the model	$1.50
4	Total cost of providing customer support per month for fraudulent transactions detected by the model (TF*$1.5)	$42.00
5	Average number of transactions per month that are fraudulent but not detected by the model (FN)	203
6	Cost incurred due to fraudulent transactions left undetected by the model (FN*c)	$1,07,590.00
7	Cost incurred per month after the model is built and deployed = (TF*$1.5) + [(FN) * (avg amt / fraud transaction)]	$1,07,632.00
8	Final savings = Cost incurred before - Cost incurred after(1-7)	$1,05,428.00

THANK
YOU